

# Security risk assessment report

## Part 1: Select up to three hardening tools and methods to implement

The following measures are recommended to address the four identified vulnerabilities:

1. Implement a password policy that mandates strong passwords and periodic password changes.
2. Use MFA for authentication.
3. Enhance firewall configuration to restrict traffic entering and leaving the network.

MFA requires users to use more than one way to identify and verify their credentials before accessing an application. Some MFA methods include fingerprint scans, ID cards, pin numbers, and passwords.

Password policies can be refined to include rules on password length, a list of acceptable characters, and a disclaimer discouraging password sharing. They can also include rules surrounding unsuccessful login attempts, such as the user losing access to the network after five unsuccessful attempts.

Firewall maintenance entails checking and updating security configurations regularly to stay ahead of potential threats.

## Part 2: Explain your recommendations

Enhancing password security helps reduce the likelihood of password leaks and successful brute-force attacks. Implementing Multi-Factor Authentication (MFA) significantly lowers the risk of identity impersonation and account takeover. Improved firewall configurations enable better monitoring of inbound and outbound traffic, allowing for the immediate blocking of suspicious activity.

Enforcing multi-factor authentication (MFA) adds an extra layer of security beyond just passwords. Because authentication requires multiple methods, it becomes more difficult for attackers to breach the network using brute-force or similar tactics, thereby reducing the likelihood of such attacks succeeding. MFA also helps curb password sharing; since the recipient of a shared

password would still need additional authentication credentials to gain access, the value of sharing a password diminishes, leading to fewer instances of the practice.

Establishing and enforcing internal password policies makes it harder for attackers to compromise the network. For example, policies that suspend accounts after multiple failed login attempts effectively thwart brute-force attacks. Mandating password complexity, requiring frequent updates, and prohibiting the reuse of old passwords also help prevent attackers from infiltrating the network.

Firewall maintenance should be performed regularly. Network administrators must ensure that firewall rules align with the latest security standards regarding traffic allowance and denial. Traffic originating from suspicious sources should be blocked. Firewall rules should be promptly updated whenever a security incident occurs—particularly those involving suspicious traffic entering the network. Implementing these measures helps defend against various DoS (Denial of Service) and DDoS (Distributed Denial of Service) attacks.